

# Predicting treatment delays of more than 90 days in prostate cancer: added value of county rurality, county income and marital status over clinical characteristics

A machine learning cohort study of 330,827 men treated with surgery or radiotherapy in US SEER registries, 2010 to 2022

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## How to read this supplement

- **Source of tables.** Every table is generated by script from the analysis reports in the public repository, so the values match the analysis outputs. The reports, in the `reports` folder, contain further tables not reproduced here, including results for every model step, leave-one-variable-out refits, bounds by income and marital status, and the protocol amendment log.
- **Skill and intervals.** Skill added is in percentage points of out-of-fold log-loss skill. Intervals are 95% cluster bootstrap intervals over rurality by county income cells (500 resamples of out-of-fold predictions, no refitting) unless stated, so they exclude model-fitting variability.
- **Disclosure control.** As the SEER Research Data Use Agreement requires, statistics resting on 1 to 4 men are shown as <5, and numbers of men in cross-tabulations are rounded to the nearest 10.
- **Later robustness analyses.** Tables S3 and S4 and the indices by period in Table S7 come from analyses added by protocol amendment 1.9 and specified before they were run. The two-model agreement rule for standardised contrasts was adopted post hoc (amendment 1.7).

## Box S1. Survival comparisons and immortal time bias (not estimated)

Comparing survival between treated and untreated men is exposed to immortal time bias, because treated men must survive until treatment starts. A simulation study of observational research found that time-fixed and exclusion methods overestimated a treatment's benefit, that a 1-year landmark method reduced the bias but did not remove it, and recommended the time-dependent method [1]. A suitable design would treat treatment as a time-dependent exposure, with competing risks of prostate cancer death and death from other causes, and use a 12-month landmark analysis only as a sensitivity analysis. This design is described, not estimated, in this study.

## Box S2. Candidate Australian analogues of the SEER variables (no Australian data analysed)

Published analyses of the Prostate Cancer Outcomes Registry in Tasmania used the measures below [2, 3]. They are candidate analogues, not equivalents: the geographic units, the construction of the measures and the health systems differ, and availability for any new study must be confirmed against the registry's data dictionary.

- **Rurality.** The county Rural-Urban Continuum Code has a candidate analogue in Australian Statistical Geography Standard remoteness areas, assigned by residential postcode.
- **Area income.** County median household income has a candidate analogue in the SEIFA Index of Relative Socio-Economic Advantage and Disadvantage, assigned by postcode.
- **Insurance and sector.** SEER has no insurance variable. The closest Australian contrast studied is a public or private treating facility.

Data gaps reported in those analyses were no comorbidity data [2], patient-reported outcomes collected consistently only from 2018 [3], and no way to tell whether external beam radiotherapy was given in a public or private facility [3]. Referral, biopsy and multidisciplinary meeting dates are not assumed to be available.

## Supplementary tables

**Table S1.** Crude waiting beyond 90 days by risk group and county rurality

| Risk group   | County rurality                 | Men    | Waited more than 90 days (%) | Median days | 90th percentile days |
|--------------|---------------------------------|--------|------------------------------|-------------|----------------------|
| Low          | Metro, 1 million or more        | 34,260 | 50.7                         | 91          | 217                  |
|              | Metro, 250,000 to 1 million     | 11,930 | 47.0                         | 86          | 211                  |
|              | Metro, under 250,000            | 4,780  | 42.3                         | 82          | 198.2                |
|              | Nonmetro, adjacent to metro     | 3,990  | 39.0                         | 77          | 187                  |
|              | Nonmetro, not adjacent to metro | 2,510  | 39.0                         | 76          | 177                  |
|              | Unknown                         | 20     | 34.8                         | 81          | 122.8                |
| Intermediate | Metro, 1 million or more        | 75,190 | 45.5                         | 84          | 182                  |
|              | Metro, 250,000 to 1 million     | 28,140 | 41.1                         | 79          | 172                  |
|              | Metro, under 250,000            | 10,070 | 36.7                         | 75          | 165                  |
|              | Nonmetro, adjacent to metro     | 8,830  | 36.1                         | 74          | 161                  |
|              | Nonmetro, not adjacent to metro | 5,360  | 36.1                         | 71          | 160                  |
|              | Unknown                         | 40     | 42.5                         | 71          | 200.5                |
| High         | Metro, 1 million or more        | 75,760 | 35.7                         | 73          | 157                  |
|              | Metro, 250,000 to 1 million     | 29,320 | 30.0                         | 67          | 142                  |
|              | Metro, under 250,000            | 10,130 | 27.9                         | 65          | 140                  |
|              | Nonmetro, adjacent to metro     | 9,590  | 27.7                         | 63          | 134.5                |
|              | Nonmetro, not adjacent to metro | 6,010  | 26.0                         | 62          | 133                  |
|              | Unknown                         | 90     | 28.4                         | 61.5        | 141.5                |
| Unknown      | Metro, 1 million or more        | 9,460  | 44.2                         | 82          | 224                  |
|              | Metro, 250,000 to 1 million     | 2,770  | 42.2                         | 80          | 203                  |
|              | Metro, under 250,000            | 1,120  | 35.0                         | 70          | 179                  |
|              | Nonmetro, adjacent to metro     | 950    | 36.5                         | 72          | 174.2                |
|              | Nonmetro, not adjacent to metro | 510    | 29.6                         | 63.5        | 162                  |
|              | Unknown                         | 10     | <5                           | 77          | 212.7                |

Crude values; nothing is adjusted. Numbers of men are rounded to the nearest 10. Medians and 90th percentiles count top-coded intervals as 731 days. <5: statistic suppressed because 1 to 4 men in the row did, or did not, wait more than 90 days.

**Table S2.** Performance at step 2 by risk group and model

| Stratum           | Model    | Log-loss skill (%) | Brier skill (%) | AUC   | Calibration intercept | Calibration slope |
|-------------------|----------|--------------------|-----------------|-------|-----------------------|-------------------|
| All men           | Logistic | 3.63               | 4.66            | 0.638 | −0.000                | 1.000             |
|                   | LightGBM | 4.17               | 5.34            | 0.648 | −0.000                | 1.030             |
| Low risk          | Logistic | 0.76               | 1.03            | 0.592 | 0.000                 | 0.988             |
|                   | LightGBM | 0.99               | 1.35            | 0.604 | −0.000                | 0.937             |
| Intermediate risk | Logistic | 1.06               | 1.40            | 0.605 | −0.000                | 1.001             |
|                   | LightGBM | 1.48               | 1.96            | 0.615 | −0.000                | 1.008             |
| High risk         | Logistic | 4.39               | 5.42            | 0.654 | 0.000                 | 1.001             |
|                   | LightGBM | 4.90               | 6.04            | 0.663 | −0.000                | 1.023             |
| Unknown risk      | Logistic | 1.07               | 1.43            | 0.610 | 0.000                 | 0.982             |
|                   | LightGBM | 1.96               | 2.64            | 0.629 | −0.000                | 0.882             |

Out-of-fold predictions at step 2 (clinical need plus area and marital characteristics). Skill is against the step 0 model. Calibration intercept: 0 is ideal; slope: 1 is ideal, below 1 means predictions are too extreme. Values for every step are in `reports/phase4_models.md`.

**Table S3.** Robustness of the area and marital increment to tuning, fold assignment and clustering

| Stratum           | Model    | Primary analysis (tuned at step 3) | Tuned at each step  | Mean (range) over 10 fold seeds | Clustered by county income band only |
|-------------------|----------|------------------------------------|---------------------|---------------------------------|--------------------------------------|
| All men           | Logistic | 0.62 (0.39 to 0.87)                | 0.62 (0.39 to 0.87) | 0.62 (0.61 to 0.62)             | 0.62 (0.38 to 0.90)                  |
|                   | LightGBM | 0.99 (0.73 to 1.30)                | 0.99 (0.73 to 1.30) | 0.99 (0.98 to 0.99)             | 0.99 (0.72 to 1.35)                  |
| Low risk          | Logistic | 0.62 (0.32 to 0.94)                | 0.62 (0.32 to 0.94) | 0.62 (0.59 to 0.63)             | 0.62 (0.28 to 1.07)                  |
|                   | LightGBM | 0.96 (0.67 to 1.28)                | 0.96 (0.67 to 1.28) | 0.98 (0.95 to 1.01)             | 0.96 (0.64 to 1.36)                  |
| Intermediate risk | Logistic | 0.56 (0.35 to 0.83)                | 0.56 (0.34 to 0.83) | 0.56 (0.54 to 0.56)             | 0.56 (0.33 to 0.89)                  |
|                   | LightGBM | 0.93 (0.65 to 1.27)                | 0.93 (0.65 to 1.27) | 0.93 (0.91 to 0.97)             | 0.93 (0.63 to 1.31)                  |
| High risk         | Logistic | 0.86 (0.60 to 1.15)                | 0.86 (0.60 to 1.15) | 0.85 (0.85 to 0.86)             | 0.86 (0.58 to 1.15)                  |
|                   | LightGBM | 1.18 (0.90 to 1.56)                | 1.18 (0.90 to 1.54) | 1.18 (1.16 to 1.20)             | 1.18 (0.87 to 1.55)                  |
| Unknown risk      | Logistic | 0.42 (0.08 to 0.80)                | 0.40 (0.07 to 0.80) | 0.41 (0.32 to 0.48)             | 0.40 (0.07 to 0.78)                  |
|                   | LightGBM | 1.37 (0.80 to 2.19)                | 1.37 (0.80 to 2.19) | 1.29 (1.22 to 1.43)             | 1.37 (0.78 to 2.09)                  |

Percentage points of out-of-fold log-loss skill added by area and marital characteristics (step 1 to 2), with 95% cluster bootstrap intervals (first fold seed, 500 resamples, no refitting). Tuning was not repeated per seed. Added by protocol amendment 1.9 and specified before being run.

**Table S4.** Stage-free clinical block and year of diagnosis as categories

| Analysis                  | Stratum           | Model    | Clinical need added<br>(95% interval) | Area and marital added<br>(95% interval) |
|---------------------------|-------------------|----------|---------------------------------------|------------------------------------------|
| Stage-free clinical block | All men           | Logistic | 2.97 (2.64 to 3.34)                   | 0.62 (0.39 to 0.87)                      |
|                           |                   | LightGBM | 3.13 (2.80 to 3.48)                   | 1.02 (0.75 to 1.33)                      |
|                           | Low risk          | Logistic | 0.14 (0.08 to 0.20)                   | 0.62 (0.32 to 0.94)                      |
|                           |                   | LightGBM | 0.02 (−0.09 to 0.13)                  | 0.96 (0.67 to 1.28)                      |
|                           | Intermediate risk | Logistic | 0.49 (0.41 to 0.57)                   | 0.56 (0.34 to 0.83)                      |
|                           |                   | LightGBM | 0.55 (0.44 to 0.65)                   | 0.93 (0.65 to 1.27)                      |
|                           | High risk         | Logistic | 3.42 (3.07 to 3.76)                   | 0.85 (0.59 to 1.14)                      |
|                           |                   | LightGBM | 3.64 (3.27 to 3.99)                   | 1.19 (0.91 to 1.57)                      |
|                           | Unknown risk      | Logistic | 0.66 (0.41 to 0.92)                   | 0.40 (0.06 to 0.80)                      |
|                           |                   | LightGBM | 0.59 (0.27 to 0.95)                   | 1.37 (0.80 to 2.19)                      |
| Year as categories        | All men           | Logistic | 2.98 (2.66 to 3.34)                   | 0.62 (0.39 to 0.86)                      |
|                           | Low risk          | Logistic | 0.15 (0.09 to 0.21)                   | 0.60 (0.31 to 0.92)                      |
|                           | Intermediate risk | Logistic | 0.50 (0.42 to 0.58)                   | 0.56 (0.34 to 0.83)                      |
|                           | High risk         | Logistic | 3.48 (3.12 to 3.82)                   | 0.87 (0.61 to 1.16)                      |
|                           | Unknown risk      | Logistic | 0.65 (0.41 to 0.90)                   | 0.47 (0.11 to 0.89)                      |

Stage-free: summary stage removed from the clinical block, risk strata unchanged. Year as categories: logistic regression only, one indicator per year replacing linear year and the 2020 indicator. Both use per-step tuning and the first fold seed. Primary estimates are in Table 2 of the main text. Added by protocol amendment 1.9 and specified before being run.

**Table S5.** Sensitivity analyses

| Scenario                                        | Threshold<br>(days) | Men     | Delayed<br>(%) | Added, logistic<br>(95% interval) | Added, LightGBM<br>(95% interval) | Area<br>contrast         | Never married<br>contrast |
|-------------------------------------------------|---------------------|---------|----------------|-----------------------------------|-----------------------------------|--------------------------|---------------------------|
| Primary analysis                                | 90                  | 330,830 | 39.7           | 0.62 (0.39 to 0.87)               | 0.99 (0.73 to 1.30)               | −9.2 / −8.8              | 6.7 / 5.9                 |
| Threshold of 60 days                            | 60                  | 330,830 | 66.6           | 0.54 (0.31 to 0.79)               | 0.94 (0.73 to 1.18)               | −10.6 / −10.3            | 3.7 / 3.2                 |
| Threshold of 120 days                           | 120                 | 330,830 | 23.0           | 0.68 (0.45 to 0.93)               | 1.06 (0.79 to 1.38)               | −6.9 / −7.1              | 6.0 / 5.0                 |
| Threshold of 180 days                           | 180                 | 330,830 | 9.1            | 0.66 (0.41 to 0.92)               | 1.00 (0.69 to 1.35)               | −3.0 / −3.2 <sup>a</sup> | 3.4 / 3.1                 |
| Risk groups from Gleason score and PSA only     | 90                  | 330,830 | 39.7           | 0.62 (0.39 to 0.87)               | 0.99 (0.73 to 1.30)               | −9.2 / −8.8              | 6.7 / 5.9                 |
| Radical prostatectomy without radiotherapy only | 90                  | 159,580 | 41.1           | 0.66 (0.48 to 0.88)               | 1.00 (0.79 to 1.26)               | −8.9 / −7.8              | 7.1 / 6.9                 |
| Excluding men diagnosed in 2020                 | 90                  | 306,960 | 39.6           | 0.64 (0.40 to 0.91)               | 1.03 (0.75 to 1.35)               | −9.4 / −8.9              | 6.8 / 5.9                 |
| Including men diagnosed in 2023                 | 90                  | 358,540 | 40.8           | 0.56 (0.35 to 0.80)               | 0.94 (0.69 to 1.22)               | −8.6 / −8.3              | 6.4 / 5.8                 |
| Including intervals of 0 days                   | 90                  | 338,580 | 38.8           | 0.61 (0.38 to 0.86)               | 0.97 (0.70 to 1.27)               | −9.2 / −9.2              | 6.5 / 5.8                 |
| One primary cancer only (strict first primary)  | 90                  | 296,120 | 40.1           | 0.61 (0.38 to 0.85)               | 0.98 (0.73 to 1.27)               | −9.1 / −9.3              | 6.6 / 5.6                 |
| Excluding prostatectomy not otherwise specified | 90                  | 330,180 | 39.7           | 0.62 (0.39 to 0.87)               | 0.98 (0.72 to 1.28)               | −9.2 / −9.1              | 6.7 / 5.9                 |

All men. Each scenario changes one setting from the primary analysis. Added: percentage points of log-loss skill added by area and marital characteristics. Contrasts are standardised differences in percentage points (logistic / LightGBM): nonmetro counties not adjacent to a metro area minus metro areas of 1 million or more, each at its typical county income; and never married minus married. <sup>a</sup>Two-model rule (both 3 points or more, same direction) not met. Threshold scenarios use a different outcome, so their values are not directly comparable. Numbers of men are rounded to the nearest 10.

**Table S6.** Standardised contrasts in waiting more than 90 days, by risk group

| Stratum           | Contrast                                            | Logistic | LightGBM | Two-model rule met |
|-------------------|-----------------------------------------------------|----------|----------|--------------------|
| All men           | Observed minus reference profile                    | −1.1     | −1.6     | no, both under 3   |
|                   | Nonmetro not adjacent minus metro 1 million or more | −9.2     | −8.8     | yes                |
|                   | Never married minus married                         | 6.7      | 5.9      | yes                |
| Low risk          | Observed minus reference profile                    | −3.4     | −4.7     | yes                |
|                   | Nonmetro not adjacent minus metro 1 million or more | −11.3    | −13.5    | yes                |
|                   | Never married minus married                         | 3.6      | 2.4      | no, one model only |
| Intermediate risk | Observed minus reference profile                    | −1.4     | −1.6     | no, both under 3   |
|                   | Nonmetro not adjacent minus metro 1 million or more | −8.7     | −9.5     | yes                |
|                   | Never married minus married                         | 6.4      | 5.5      | yes                |
| High risk         | Observed minus reference profile                    | 0.1      | −0.1     | no, both under 3   |
|                   | Nonmetro not adjacent minus metro 1 million or more | −8.4     | −7.2     | yes                |
|                   | Never married minus married                         | 8.3      | 7.6      | yes                |
| Unknown risk      | Observed minus reference profile                    | −0.1     | 0.0      | no, both under 3   |
|                   | Nonmetro not adjacent minus metro 1 million or more | −12.1    | −14.2    | yes                |
|                   | Never married minus married                         | 6.0      | 6.2      | yes                |

Standardised percentage points waiting more than 90 days (g-computation with the step 2 model; each man keeps his own clinical features and year). Reference profile: married, metro area of 1 million or more, county income rank at the 83.33% quantile. Area profiles set rurality and county income together at the median income band of men living in that type of area. Rule (post hoc, amendment 1.7): both models 3 points or more in the same direction. No intervals are reported.

**Table S7.** Income concentration of delay, overall and by period of diagnosis

| Stratum           | 2010 to 2022           | 2010 to 2014            | 2015 to 2019            | 2020 to 2022            |
|-------------------|------------------------|-------------------------|-------------------------|-------------------------|
| All men           | 0.062 (0.039 to 0.083) | 0.059 (0.025 to 0.092)  | 0.037 (0.011 to 0.065)  | 0.027 (0.003 to 0.061)  |
| Low risk          | 0.103 (0.068 to 0.134) | 0.092 (0.053 to 0.136)  | 0.084 (0.052 to 0.121)  | 0.052 (0.017 to 0.092)  |
| Intermediate risk | 0.075 (0.051 to 0.096) | 0.058 (0.023 to 0.099)  | 0.048 (0.019 to 0.080)  | 0.039 (0.015 to 0.067)  |
| High risk         | 0.041 (0.017 to 0.065) | 0.028 (−0.011 to 0.062) | 0.020 (−0.004 to 0.049) | 0.015 (−0.013 to 0.056) |
| Unknown risk      | 0.077 (0.037 to 0.128) | 0.060 (0.001 to 0.130)  | 0.023 (−0.041 to 0.100) | 0.049 (−0.005 to 0.101) |

Erreygers-corrected concentration index (95% cluster bootstrap interval) of waiting more than 90 days, ranking men by county median household income, poorest first. Positive values mean delay is concentrated in higher-income counties. Crude: not standardised for clinical features, and it does not separate income from rurality. Indices by period were added by protocol amendment 1.9 and specified before being run.

**Table S8.** Men without a recorded interval: bounds and inverse probability weighting

| County rurality                 | Treated men | Recorded men (%) | Lower bound (%) | Upper bound (%) | No recorded interval (%) | Weighted (%) |
|---------------------------------|-------------|------------------|-----------------|-----------------|--------------------------|--------------|
| Metro, 1 million or more        | 205,980     | 42.5             | 40.2            | 45.7            | 5.5                      | 42.3         |
| Metro, 250,000 to 1 million     | 75,790      | 37.6             | 35.8            | 40.6            | 4.8                      | 37.4         |
| Metro, under 250,000            | 27,050      | 34.2             | 33.0            | 36.6            | 3.6                      | 34.1         |
| Nonmetro, adjacent to metro     | 24,050      | 33.2             | 32.2            | 35.1            | 2.9                      | 33.1         |
| Nonmetro, not adjacent to metro | 14,780      | 32.2             | 31.3            | 34.0            | 2.7                      | 32.0         |
| Unknown                         | 170         | 33.3             | 31.9            | 36.1            | 4.2                      | 33.5         |

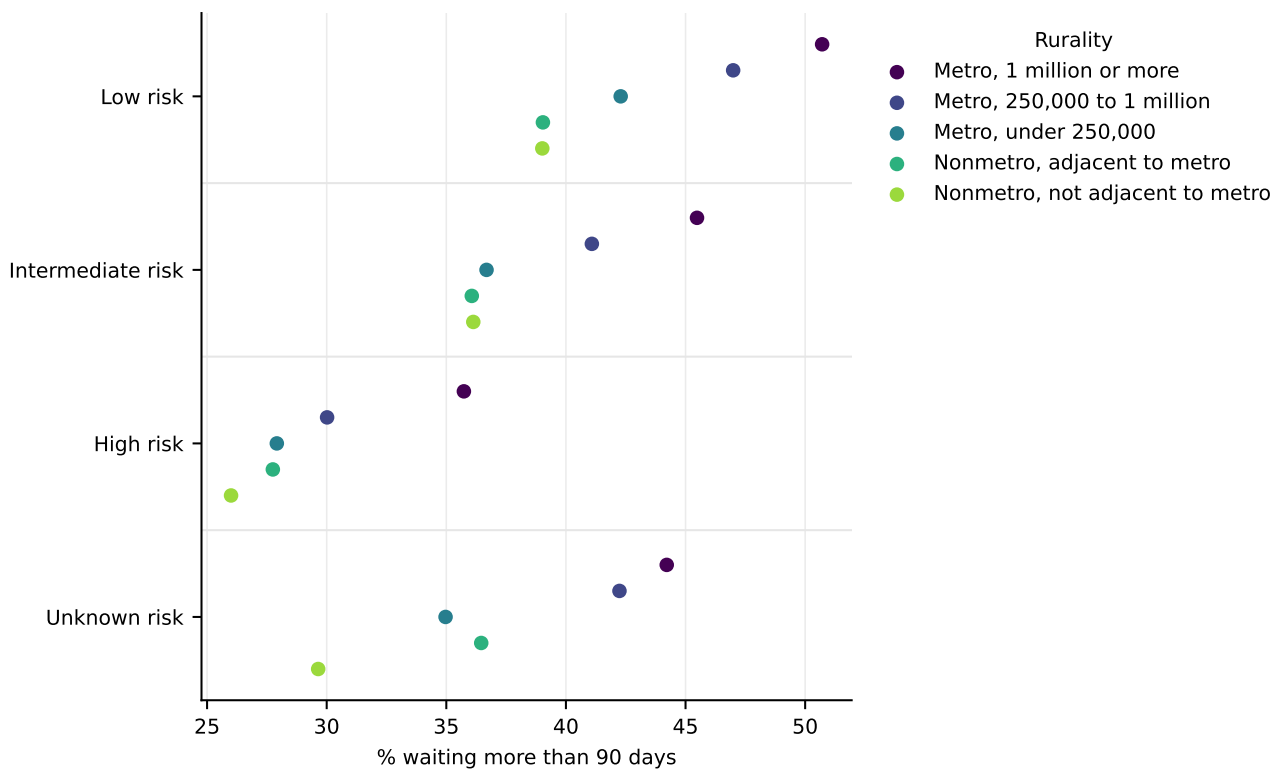
Percentage waiting more than 90 days among treated men meeting cohort steps 1 to 6 (intervals of 0 days excluded). Lower bound: every man without a recorded interval waited 90 days or less; upper bound: every such man waited longer. Weighted: men with a recorded interval, weighted by the inverse of their predicted probability of having one. Numbers of men rounded to the nearest 10. Neither approach addresses selection into recorded treatment.

**Table S9.** Secondary outcome: recorded radical prostatectomy or radiotherapy

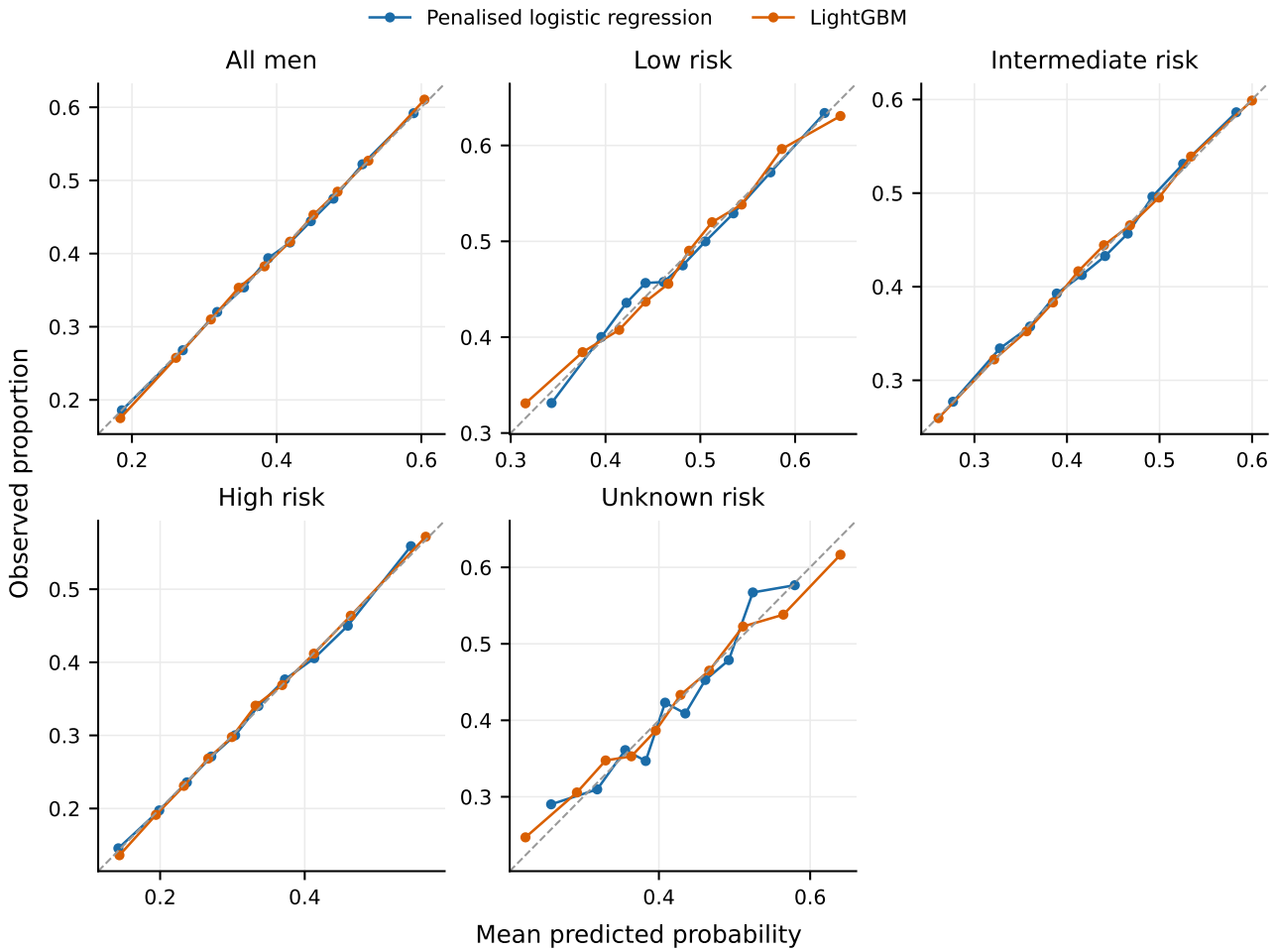
| Stratum                    | Model    | Skill at step 2 (%) | Clinical need added (95% interval) | Area and marital added (95% interval) | AUC at step 2 | Area contrast | Never married contrast |
|----------------------------|----------|---------------------|------------------------------------|---------------------------------------|---------------|---------------|------------------------|
| Intermediate and high risk | Logistic | 17.18               | 15.22 (14.85 to 15.65)             | 1.96 (1.72 to 2.31)                   | 0.771         | −2.9          | −6.5                   |
|                            | LightGBM | 20.49               | 18.32 (17.92 to 18.77)             | 2.16 (1.92 to 2.54)                   | 0.791         | −1.4          | −5.9                   |
| Intermediate risk          | Logistic | 8.79                | 6.81 (6.38 to 7.21)                | 1.97 (1.65 to 2.42)                   | 0.692         | −2.2          | −7.2                   |
|                            | LightGBM | 10.14               | 7.83 (7.42 to 8.20)                | 2.31 (1.99 to 2.82)                   | 0.703         | −0.3          | −6.4                   |
| High risk                  | Logistic | 29.29               | 27.25 (26.63 to 27.96)             | 2.04 (1.84 to 2.33)                   | 0.854         | −3.3          | −5.8                   |
|                            | LightGBM | 31.76               | 29.54 (28.94 to 30.17)             | 2.22 (2.00 to 2.53)                   | 0.865         | −2.5          | −5.1                   |

Outcome: a record of radical prostatectomy or radiotherapy in the first course, among intermediate- and high-risk men meeting every cohort step before treatment. No record can mean active surveillance, watchful waiting, hormone therapy only, refusal or uncaptured treatment, so a lower percentage cannot be read as under-treatment. Contrasts are standardised percentage points, defined as in Table S6.

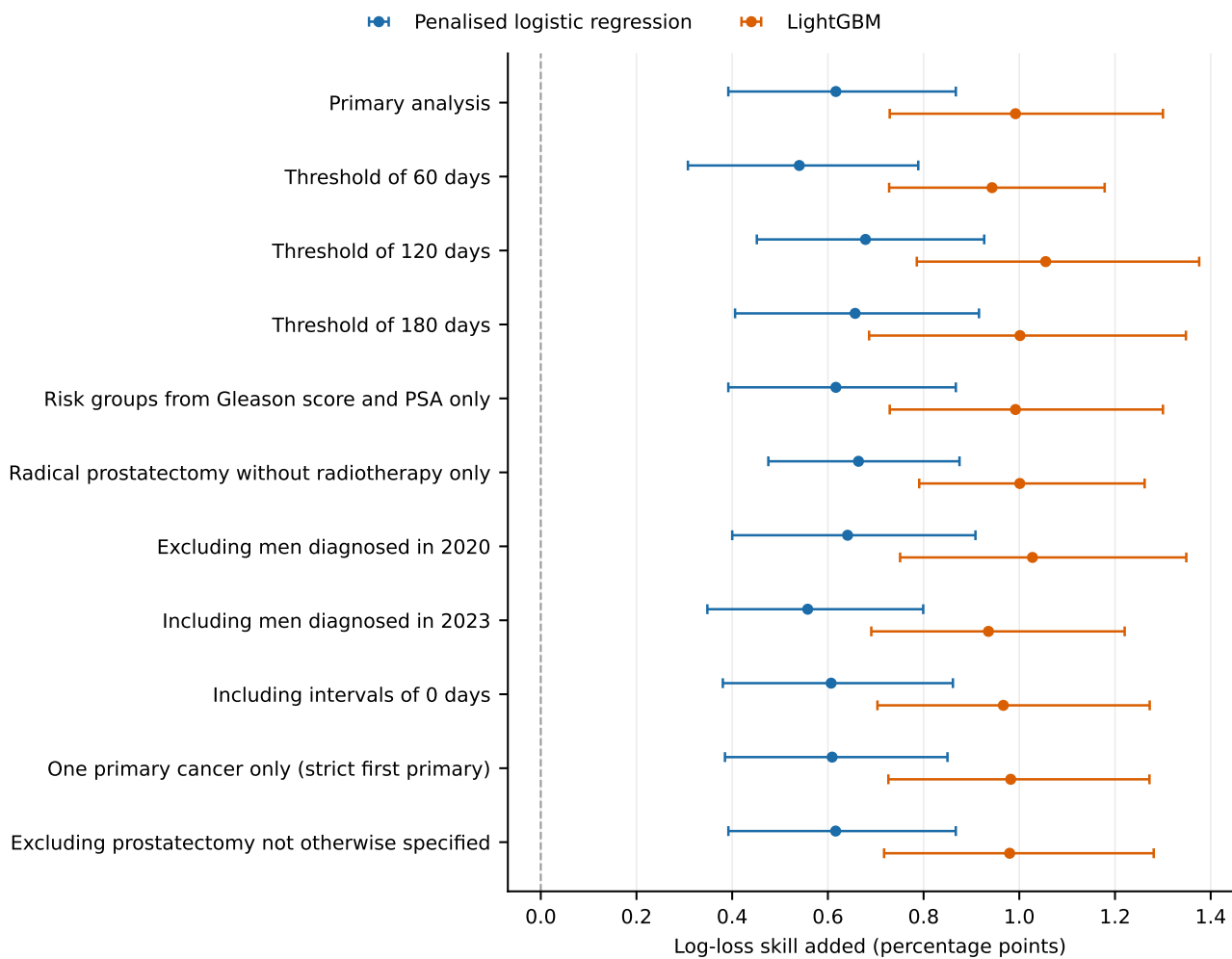
Supplementary figures



**Figure S1.** Crude percentage of men waiting more than 90 days from diagnosis to first recorded treatment, by risk group and county rurality. Men with unknown rurality, and percentages resting on 1 to 4 men, are not shown. Nothing is adjusted.



**Figure S2.** Calibration of out-of-fold predicted probabilities at step 2 (clinical need plus area and marital characteristics), in 10 equal-count bins, by risk group and model type. The dashed line marks perfect calibration.



**Figure S3.** Log-loss skill added by the area and marital block under each sensitivity scenario, all men, with 95% cluster bootstrap intervals. Each scenario changes one setting from the primary analysis. The threshold scenarios use a different outcome, so their values are not directly comparable with the others.

## Reporting checklists

**Table S10.** STROBE and RECORD checklist. Item wording is abbreviated from the RECORD statement, which includes the STROBE items [4, 5]. Locations are sections of the main text unless marked S (this supplement).

| Item                    | Topic                                             | Where reported                                                                                                                               |
|-------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1a, 1b                  | Design in title or abstract; informative abstract | Abstract (retrospective cohort study)                                                                                                        |
| RECORD 1.1 to 1.3       | Database, region and time frame; linkage          | Title; Abstract; Methods, Data and cohort (county attributes linked by SEER; no person-level linkage)                                        |
| 2, 3                    | Background; objectives                            | Introduction                                                                                                                                 |
| 4, 5                    | Study design; setting and dates                   | Methods, Data and cohort                                                                                                                     |
| 6a                      | Eligibility and selection                         | Methods, Data and cohort; Figure 1                                                                                                           |
| 6b                      | Matching                                          | Not applicable                                                                                                                               |
| RECORD 6.1 to 6.3       | Selection codes; their validation; linkage flow   | Methods, Data and cohort (codes in <code>config/analysis.yaml</code> ); no validation study of the cohort algorithm; no person-level linkage |
| 7, RECORD 7.1           | Outcomes and predictors; complete list of codes   | Methods, Outcome and risk groups, and Ordered feature steps; <code>config/analysis.yaml</code>                                               |
| 8                       | Data sources and measurement                      | Methods, Data and cohort, and Outcome and risk groups                                                                                        |
| 9                       | Efforts to address bias                           | Methods, Robustness, selection and secondary outcome; Discussion (limitations)                                                               |
| 10                      | Study size                                        | Methods, Data and cohort (all eligible men); Figure 1                                                                                        |
| 11                      | Quantitative variables                            | Methods, Ordered feature steps                                                                                                               |
| 12a, 12b                | Statistical methods; subgroups and interactions   | Methods, Models and validation (risk-group strata; clinical interaction terms), and Standardised percentages and income concentration        |
| 12c                     | Missing data                                      | Methods, Ordered feature steps, Models and validation, and Robustness, selection and secondary outcome (bounds and weighting); Table S8      |
| 12d                     | Loss to follow-up                                 | Not applicable (top-coded intervals counted as delayed)                                                                                      |
| 12e                     | Sensitivity analyses                              | Methods, Robustness, selection and secondary outcome; Tables S3 to S5                                                                        |
| RECORD 12.1 to 12.3     | Database access; data cleaning; linkage           | Full case listing of the selected site and years; every export label mapped explicitly in code; no person-level linkage                      |
| 13a to 13c, RECORD 13.1 | Numbers at each stage; flow diagram               | Results, Cohort and delay; Figure 1                                                                                                          |
| 14a, 14b                | Characteristics; missing data                     | Table 1 (unknown categories shown)                                                                                                           |
| 14c                     | Follow-up time                                    | Not applicable                                                                                                                               |
| 15                      | Outcome data                                      | Results, Cohort and delay; Table S1                                                                                                          |
| 16a                     | Estimates and precision                           | Table 2 (intervals); standardised contrasts without intervals, stated in Results; Table S6                                                   |
| 16b, 16c                | Category boundaries; absolute measures            | Methods, Outcome and risk groups; Table 1; Results, Standardised differences and income concentration                                        |
| 17                      | Other analyses                                    | Results, Robustness, selection and secondary outcome; Tables S3 to S9                                                                        |
| 18, 20                  | Key results; interpretation                       | Discussion                                                                                                                                   |
| 19, RECORD 19.1         | Limitations, including routinely collected data   | Discussion, limitations paragraph                                                                                                            |
| 21                      | Generalisability                                  | Discussion                                                                                                                                   |
| 22                      | Funding                                           | Declarations                                                                                                                                 |
| RECORD 22.1             | Access to protocol, data and code                 | Methods, Protocol, reporting and disclosure; Declarations                                                                                    |

**Table S11.** TRIPOD+AI checklist. Item wording is abbreviated from the TRIPOD+AI checklist [6]. Development and internal evaluation use the same data through cross-fitting; the models are measurement tools and are not proposed for clinical use.

| Item             | Topic                                                     | Where reported                                                                                        |
|------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1, 2             | Title; abstract                                           | Title; Abstract                                                                                       |
| 3a to 3c, 4      | Context; intended purpose; known inequalities; objectives | Introduction; Methods, Protocol, reporting and disclosure (not for clinical use)                      |
| 5a, 5b, 6a to 6c | Data sources, dates, setting, eligibility, treatments     | Methods, Data and cohort, and Ordered feature steps (treatment type at step 3)                        |
| 7                | Data preparation                                          | Code in the repository                                                                                |
| 8a to 8c         | Outcome; assessors; blinding                              | Methods, Outcome and risk groups; assessors and blinding not applicable (registry-abstracted outcome) |
| 9a to 9c         | Predictors                                                | Methods, Ordered feature steps (pre-specified in the protocol); assessors not applicable              |

| Item               | Topic                                                                                  | Where reported                                                                                                       |
|--------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 10, 11             | Sample size; missing data                                                              | Methods, Data and cohort (all eligible men), Ordered feature steps, and Models and validation                        |
| 12a to 12c         | Data use; predictor handling; model type, tuning and internal validation               | Methods, Models and validation (5-fold cross-fitting)                                                                |
| 12d, 23b           | Heterogeneity across clusters                                                          | Cluster bootstrap only; heterogeneity across clusters not examined                                                   |
| 12e                | Performance measures                                                                   | Methods, Models and validation; Table S2; Figure S2                                                                  |
| 12f, 24            | Model updating                                                                         | Not applicable                                                                                                       |
| 12g                | How predictions were calculated                                                        | Code in the repository                                                                                               |
| 13                 | Class imbalance                                                                        | No class imbalance methods used                                                                                      |
| 14                 | Fairness                                                                               | Not addressed as a model property; area and marital differences are the measured quantity                            |
| 15                 | Model output                                                                           | Predicted probabilities; no classification threshold                                                                 |
| 16                 | Training versus evaluation differences                                                 | Not applicable (cross-fitting within the same data)                                                                  |
| 17, 18a to 18f, 19 | Ethics; funding; conflicts; protocol; registration; data and code; patient involvement | Methods, Protocol, reporting and disclosure (not registered); Declarations                                           |
| 20a, 20b           | Participant flow; characteristics                                                      | Results, Cohort and delay; Figure 1; Table 1                                                                         |
| 20c                | Comparison with development data                                                       | Not applicable                                                                                                       |
| 21                 | Participants and events per analysis                                                   | Table 1; Table 2; Table S1                                                                                           |
| 22                 | Full model specification                                                               | Not released as a clinical model; code that refits every model is in the repository                                  |
| 23a                | Performance with intervals, including subgroups                                        | Table 2 and Table S2 by risk group (AUC and calibration without intervals); not reported by area or marital subgroup |
| 25, 26             | Interpretation; limitations                                                            | Discussion                                                                                                           |
| 27a to 27c         | Implementation; next steps                                                             | Not applicable (no implementation proposed); Discussion                                                              |

## References

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- [2] Foley GR, Blizzard CL, Stokes B, et al. Urban-rural prostate cancer disparities in a regional state of Australia. *Sci Rep*. 2022;12(1):3022.
- [3] Foley GR, Blizzard CL, Skala M, et al. Prostate cancer disparities between public and private healthcare patients in Tasmania, a regional state of Australia. *Cancers (Basel)*. 2025;18(1):79.
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- [5] Benchimol EI, Smeeth L, Guttman A, et al. The REporting of studies Conducted using Observational Routinely-collected health Data (RECORD) statement. *PLoS Med*. 2015;12(10):e1001885.
- [6] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378.